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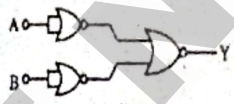
Virudhunagar District
Common Halfy Yearly Examination, - 2025

Standard 12
PHYSICS
PART - I

Time: 3.00 Hours Marks: 70

Answer all the questions. Write the suitable answer with option code: 15x1=15

1) A carbon resistor of $(47 \pm 4.7)K\Omega$ to be marked with rings of different colours for its identification. The colour code sequence will be
   a) yellow - green - violet - gold b) yellow - violet - orange - silver
   c) violet - yellow - orange - silver d) green - orange - violet - gold
2) The colours on the soap bubble when exposed to sunlight is due to
   a) Interference b) Polarisation c) diffraction d) refraction
3) The maximum power is dissipated for an alternating current in
   a) LC circuit b) pure resistive circuit
   c) a pure inductive circuit d) a pure capacitive circuit
4) The transverse nature of light is shown in
   a) Interference b) diffraction c) scattering d) polarisation
5) Atomic number of H-like atom with ionization potential 122.4V for n = 1 is
   a) 1 b) 2 c) 3 d) 4
6) S.I. unit of magnetic moment is
   a) Am^2 b) Am c) NAm^{-2} d) $NA^{-1}m^{-1}$
7) The output 'y' of the given logic gate is similar to the output of a



   a) 'OR' gate b) 'NOR' gate c) 'AND' gate d) 'NAND' gate
8) The radius of $_{30}Zn^{64}$ nucleus is
   a) 1.2 F b) 2.4 F c) 3.6 F d) 4.8 F
9) A rod of length 10 cm lies along te principal axis of a concave mirror of focal length 10 cm in such a way that its end closer to the pole is 20 cm away from the mirror. The length of the image is
   a) 2.5 cm b) 5 cm c) 10 cm d) 15 cm
10) 'Ski Wax' is an application of nano product in the field of
   a) medicine b) textile c) sports d) automotive industry
11) If $E = E_0 \sin [10^6 x - wt]$ be the electric field of a plane electromagnetic wave, the value of 'w' is
   a) 0.3×10^{-14} rad s^{-1} b) 3×10^{-14} rad s^{-1}
   c) 0.3×10^{14} rad s^{-1} d) 3×10^{14} rad s^{-1}
12) To obtain sustained oscillation in an oscillator
   a) feedback should be positive b) feedback factor must be unity
   c) phase shift must be zero (or) 2π d) all the above
13) Two radiations with photon energies 0.9eV and 3.3 eV respetively are falling on a metallic surface successively. If the function of the metal is 0.6 eV, then the ratio of maximum speeds of emitted electrons in the two cases will be
   a) 1:4 b) 1:3 c) 1:1 d) 1:9
14) An electric field $E = 10x\,i$ exists in a certain region of space. Then the potential difference $V = V_0 - V_A$, where V_0 is the potential at the origin and V_A is the potential at x = 2m is
   a) 10V b) −20V c) +20V d) −10V
15) In an oscillating LC circuit, the maximum charge on the capacitor is 'Q'. The charge on the capacitor when the energy is stared equally between the electric and magnetic fields is
   a) $Q/2$ b) $Q/\sqrt{3}$ c) $Q/\sqrt{2}$ d) Q

PART - II

Answer any six questions. Question no. 24 is compulsory: 6x2=12

16) What is meant by excitation energy?
17) Define: Temperature co-efficient of resistance.
18) State Shell's law.

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19) Define: Electric Flux. Give its unit.
20) Give two uses of infrared radiations.
21) Define: One ampere (on the basis of magnetic force)
22) A straight metal wire crosses a magnetic field of flux 4 mWb in a time of 0.4 s. Find the magnitude of the emf induced in the wire
23) In a transistor connected in the common base configuration $\alpha = 0.95$, $I_E = 1mA$, calculate the values of I_C and I_B.
24) Two light sources have intensity of light as "I_0". What is the resultant intensity at a point where the two light waves have a phase difference of $\pi/3$?

PART - III

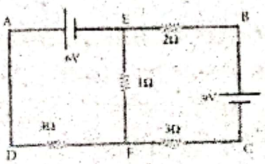
Answer any six questions. Question no. 33 is compulsory: **6x3=18**

25) Define: a) Distance closest approach b) Impact parameter
c) Mass defect of a nucleus
26) Obtain the expression for equivalent resistance when a number of resistors are connected in parallel.
27) A coil of a tangent galvanometer of diameter 0.24 m has 100 turns. If the horizontal component of Earth's magnetic field is 25×10^{-6}T, then calculate the current which gives a deflection of 60°.
28) What are the differences between polarised light and unpolarised light?
29) List out the advantages and limitations of frequency modulation.
30) Obtain the expression for self-inductance of a long solenoid.
31) Derive the expression for de-Broglie of wavelength of electrons accelerated by a potential difference of 'V' volts.
32) Light travels from air into a glass slab of thickness 50 cm and refractive index 1.5.
a) What is the speed of light in the glass slab?
b) What is the time taken by the light to travel through the glass slab?
c) What is the optical path of the glass slab?
33) A parallel plate capacitor filled with mica having $E_r = 5$ is connected to a 10V battery. The area of each parallel plate is 6 cm^2 and separation distance is 6 mm. Find the capacitance and stored charge.....

PART - IV

Answer all questions: **5x5=25**

34) a] Obtain a relation for the magnetic field at a point along the axis of a circular coil carrying current using Biot-Savart Law.
(OR)
b] Describe the Davisson - Germer experiment in detail.
35) a] Explain in detail the principle construction and working of Van de Graff Generator.
(OR)
b] Explain the Maxwell's modification of Ampere's circuital law.
36) a] i) What is half-life of a radio active nucleus? Give the expression for it.
ii) Calculate the time required for 60% of a sample of radon undergo decay. Given $T_{1/2}$ of radon = 3.8 days.
(OR)
b] Draw the circuit diagram of a half-wave rectifier and explain its working.
37) a] Explain the construction and working of a Transformer.
(OR)
b] Obtain Lens Maker's formula.
38) a] Obtain the expression for bandwidth in Young's double slit experiment.
(OR)
b] i) State the two Kirchoff's Rules.
ii) Calculate the current that flows in the 1Ω resistor in the following circuit.



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